Day 5 – Building and Verifying the T-REx Dataset Pipeline

LLM Research Log | Souvik Pal

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Task Provided

Download the T-REx dataset, understand its structure, and build a preprocessing pipeline to prepare the dataset for future relation extraction and structured reasoning experiments.

What I Did

- Downloaded the T-REx dataset from Hugging Face.
- Explored the dataset structure and understood the purpose of fields such as relation, head entity, tail entity, title, and text.
- Converted the dataset into CSV format for easier analysis.
- Built a preprocessing pipeline to map relation IDs into human-readable relation names.
- Merged and cleaned the dataset to generate processed training, validation, and test datasets.
- Verified the processed data to ensure the output was correct.
- Visualized the distribution of relation types using Matplotlib.

What I Implemented

- Dataset downloading and preprocessing pipeline.
- Relation ID to relation name mapping.
- Data verification scripts.
- Dataset merging and validation.
- Relation distribution visualization.
- Utility script to add custom test samples for future experiments.

Challenge Faced

While verifying the processed dataset, I found that the test split contained missing relation names because it used a different relation format than the training and validation splits.

How I Solved It

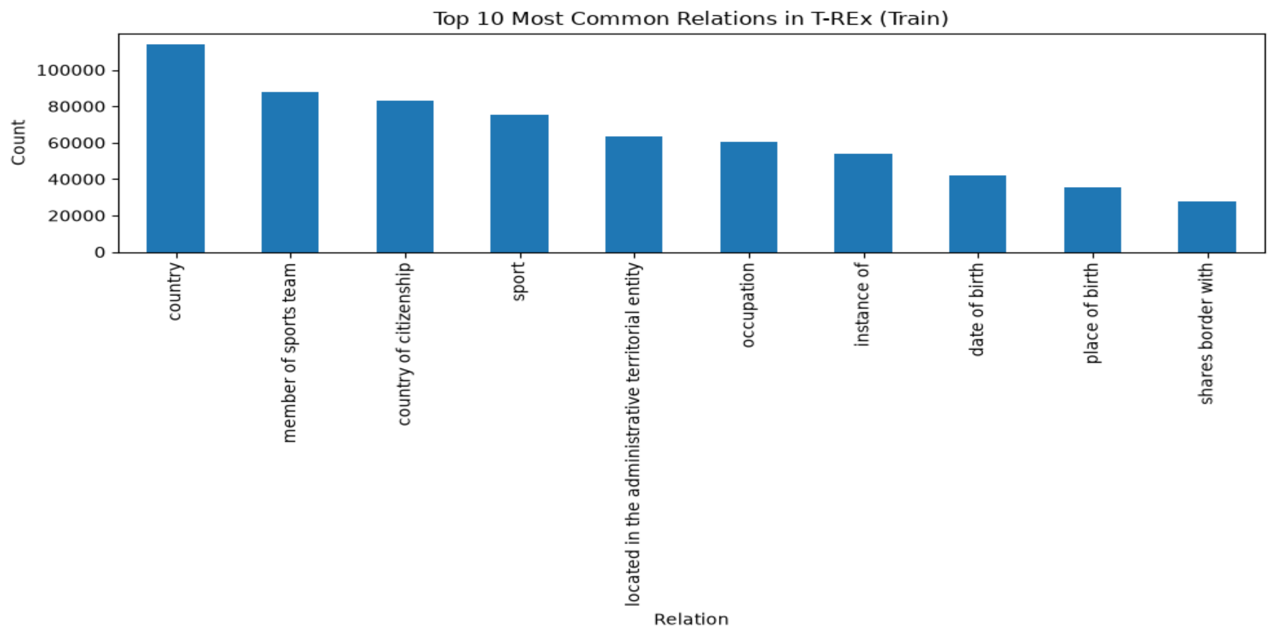
I updated the preprocessing pipeline by adding a fallback mechanism that preserved the original relation label whenever a matching relation name was unavailable. After applying this fix, all dataset splits were processed successfully without missing values.

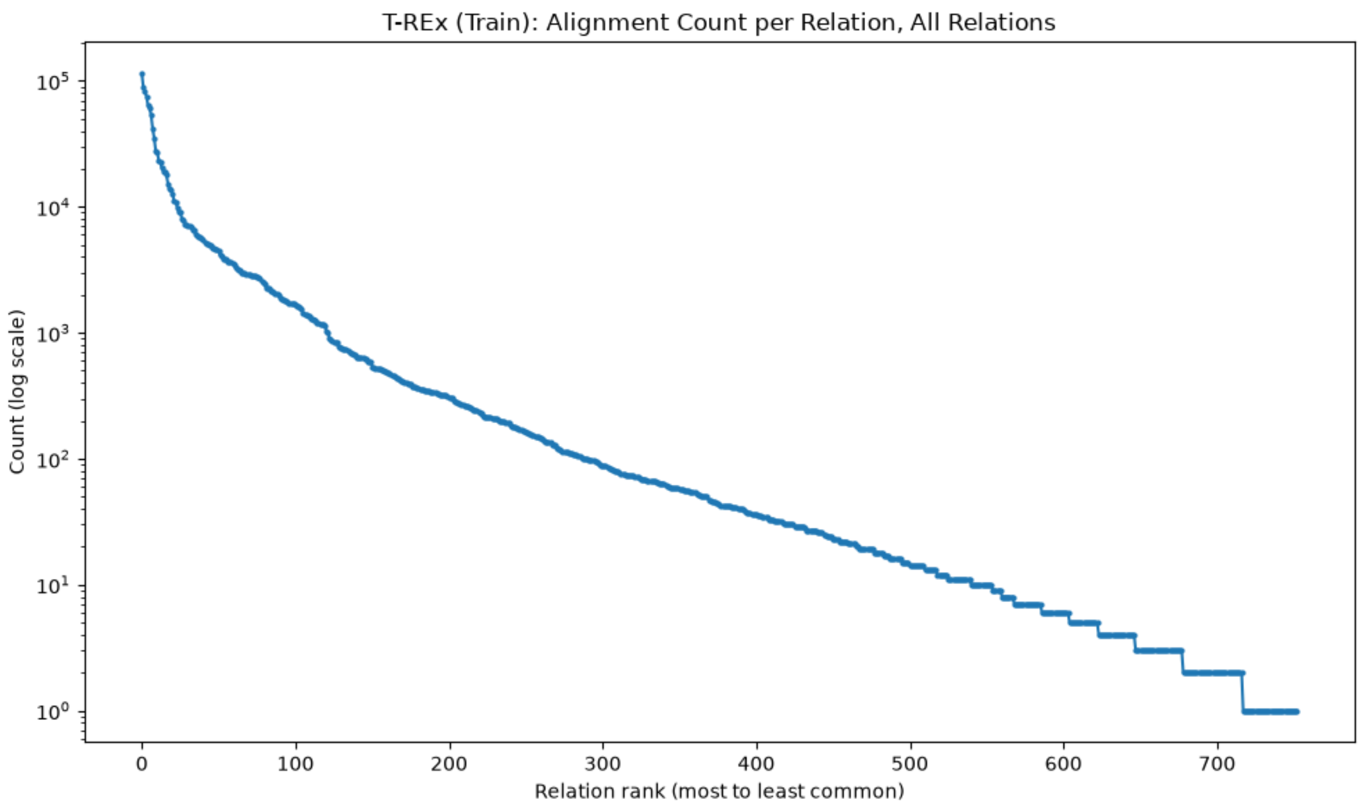
Final Output Files

File	Rows	Purpose
trex_train_sample.csv	1,274,264	Raw training dataset
trex_test_final_with_names.csv	122	Fully labeled test dataset
trex_train_final_with_names.csv	1,274,264	Fully labeled training dataset
trex_validation_final_with_names.csv	318,566	Fully labeled validation dataset
trex_test_sample.csv	122	Raw test dataset
trex_validation_sample.csv	318,566	Raw validation dataset
relation_id_to_name.csv	753 unique	Relation ID to relation name mapping
trex_test_with_my_data.csv	123	Test dataset with one custom sample

Visualization Files

- trex_top_relations.png – Top 10 most frequent relation types.
- trex_full_distribution.png – Complete relation distribution using a log-scale graph.





Results

- Successfully built and verified a complete preprocessing pipeline for the T-REx dataset.
- Generated fully labeled training, validation, and test datasets with 0 missing relation names.
- Verified 753 unique relation types between the mapping file and the processed datasets.

- Identified and fixed the relation mapping issue caused by different relation formats in the test split.
- Observed that the T-REx dataset follows a power-law distribution, where a small number of relation types dominate while many others occur infrequently.

What I Learned

- Practical understanding of the T-REx dataset and its structure.
- Importance of preprocessing and validating datasets before model training.
- Real-world datasets may contain inconsistencies that require debugging and careful verification.
- Data visualization helps identify dataset characteristics such as class imbalance and relation distribution.
- Reading about a dataset and working with it directly provide very different levels of understanding.

Conclusion

This was my first hands-on experience working with a large-scale research dataset. It helped me understand the complete workflow of downloading, preprocessing, validating, and analyzing data before using it for Large Language Model research. It also reinforced the importance of verifying outputs instead of assuming a pipeline is correct simply because it runs without errors.